Chapter 15
The Black–Scholes–Merton Model

Chapter 13 established the principle: a derivative that can be replicated has a price independent of anyone's beliefs. Chapter 14 supplied the mathematics for carrying that principle into continuous time, and ended one step short of the result. This chapter takes the step.

What emerges is a closed-form formula for European options—the most widely used equation in finance, and the one that made options a mass market rather than a specialist's trade. Its importance is not that it is right. Chapter 19 documents at length that it is not, in a specific and measurable way. Its importance is that it turned option pricing into a calculation with five inputs, of which four are read off a screen and the fifth became a traded quantity in its own right.

Sections 15.1 to 15.3 are the assumptions, the differential equation, and the formula. Sections 15.4 and 15.5 say what its terms mean and how it behaves. Sections 15.6 and 15.7 are the fifth input, volatility, estimated from history and implied from prices; Sections 15.8 to 15.10 are the formula in practice, with dividends, and for American options. Section 15.11 is what the model gets wrong and Section 15.12 where it came from.

Section 15.2 can be read for its conclusions alone; the formula in Section 15.3 is self-contained.

15.1 THE ASSUMPTIONS

The model rests on assumptions that Chapter 14 has already named. They are collected here because knowing them is how one knows when the answer will be wrong.

1. The stock follows geometric Brownian motion with constant μ and σ.
2. Short selling is permitted, with full use of the proceeds.
3. There are no transaction costs or taxes, and securities are perfectly divisible.
4. No dividends are paid over the option's life (relaxed in Section 15.9).
5. There are no arbitrage opportunities.
6. Trading is continuous.
7. The risk-free rate r is constant and the same for borrowing and lending.

Every one of these fails in reality, and the book has already said where. Assumption 1 fails through volatility clustering and jumps (Section 14.12; Chapter 19 measures what they do to option prices). Assumptions 2 and 7 fail in the way Section 5.15 documented at length: borrowing is not at r, and shorting is neither free nor always possible. Assumption 3 is what Chapter 18 is about. Assumption 6 fails whenever a market closes or a stock is suspended.

The list is not a disclaimer; it is a diagnostic tool. When a market price disagrees with the formula, the useful question is which of the seven has broken, because that identifies the repair.

15.2 THE BLACK–SCHOLES–MERTON DIFFERENTIAL EQUATION

Section 14.8 supplies the derivation. Take a stock following

$$dS = \mu S\,dt + \sigma S\,dW$$

and a derivative on it whose price at time t can be expressed as $f(S,t)$, a function of the current stock price and time alone. Itô's lemma gives

$$df = \left(\frac{\partial f}{\partial S}\mu S + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S^2\right)dt + \frac{\partial f}{\partial S}\sigma S\,dW$$

and a portfolio short one derivative and long $\partial f/\partial S$ shares has no dW term at all. Writing Π for its value,

$$\Pi = -f + \frac{\partial f}{\partial S}S, \qquad d\Pi = \left(-\frac{\partial f}{\partial t} - \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S^2\right)dt$$

The portfolio is riskless over dt, so it must earn r, or Section 13.1's arbitrage is available. Setting $d\Pi = r\Pi\,dt$ and rearranging gives the **Black–Scholes–Merton differential equation**:

$$\boxed{\frac{\partial f}{\partial t} + rS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2} = rf} \tag{15.1}$$

Three observations matter more than the derivation.

μ is absent. The stock's expected return entered through Itô's lemma and canceled when the hedge was imposed. This is Chapter 13's central result (the real-world probability is irrelevant) appearing in continuous time. Two investors who disagree completely about where the stock is going must still agree on the option's value.

The equation is the same for every derivative whose value depends only on S and t. Nothing in the derivation used the payoff, so forwards, calls, puts, digitals, and the barrier options of Chapter 34 all satisfy (15.1). What distinguishes them is the **boundary condition**, the value at $t = T$:

European call: $f = \max(S-K,0)$, forward: $f = S-K$, digital: $f = \mathbf{1}\{S > K\}$

The instrument is the boundary condition, not the equation. The qualification "only on S and t" matters. A payoff that depends on the path (an average, a running maximum) needs an extra state variable and the equation gains a term; an American option satisfies it only in the region where holding beats exercising, which the free boundary of Section 15.10 encloses; and a derivative whose value also moves with volatility or interest rates needs those as variables too. Each of those is a different equation, but the argument that produces it is this one.

The hedge is instantaneous. The portfolio is riskless over dt only, because $\partial f/\partial S$ changes as S and t change. Section 14.8 flagged this and Chapter 18 confronts its cost.

15.3 THE FORMULA

Solving (15.1) subject to the European call's boundary condition is a piece of applied mathematics carried out in [TN15], reproduced in Appendix A. The route taken there is the one Chapter 13 makes natural: rather than solving the equation directly, take the risk-neutral expectation of the payoff.

Under the risk-neutral measure, Section 14.10 gives

$$\ln S_T \sim N\left(\ln S_0 + \left(r - \frac{\sigma^2}{2}\right)T,\ \sigma^2 T\right)$$

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and taking the expected payoff $\max(S_T - K, 0)$ under that distribution and discounting it at r produces the **Black–Scholes–Merton formulas**, which the market calls simply Black–Scholes:

$$\boxed{c = S_0 N(d_1) - K e^{-rT} N(d_2)} \tag{15.2}$$

$$p = K e^{-rT} N(-d_2) - S_0 N(-d_1) \tag{15.3}$$

where $N(\cdot)$ is the standard normal cumulative distribution function and

$$d_1 = \frac{\ln(S_0/K) + \left(r + \sigma^2/2\right) T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \tag{15.4}$$

The formula takes five inputs: S_0, K, T, r, and σ. Four are observable; the fifth is not, and Sections 15.6 and 15.7 are about it.

Two routes to the same answer

There are two ways to get from (15.1) to (15.2), and both are worth recognizing because different parts of the book use different ones.

The **differential-equation route** solves (15.1) directly, subject to $f = \max(S - K, 0)$ at T. A change of variables turns it into the heat equation of physics, whose solution has been known since the nineteenth century. This is Black and Scholes's original route, and it is the one that generalizes to the numerical methods of Chapter 20.

The **risk-neutral route** takes the discounted expectation instead:

$$c = e^{-rT} \int_K^\infty (S_T - K)\, g(S_T)\, dS_T$$

where g is the lognormal density of (14.16). Splitting the integrand,

$$c = \underbrace{e^{-rT} \int_K^\infty S_T\, g\, dS_T}_{S_0 N(d_1)} - \underbrace{K e^{-rT} \int_K^\infty g\, dS_T}_{K e^{-rT} N(d_2)}$$

The second integral is $Q(S_T > K)$, the source of $N(d_2)$; the first is an expectation of S_T restricted to the exercise region, the source of $N(d_1)$. [TN15] evaluates both.

This route is the one to remember, because it makes the formula's two terms mean something (Section 15.4 unpacks them), and because it is the route that extends to every payoff in Chapter 34 without modification.

A check on the equation. It is worth confirming that (15.1) really does govern instruments other than options. Take a forward contract with delivery price K, worth $f = S - K e^{-r(T-t)}$ by Section 5.5. Then

$$\frac{\partial f}{\partial t} = -rK e^{-r(T-t)}, \qquad \frac{\partial f}{\partial S} = 1, \qquad \frac{\partial^2 f}{\partial S^2} = 0$$

and substituting into the left-hand side of (15.1),

$$-rK e^{-r(T-t)} + rS \times 1 + 0 = r\left(S - K e^{-r(T-t)}\right) = rf$$

The equation is satisfied identically. The forward contract's value from Chapter 5, derived there from a cash-and-carry argument with no model at all, is a solution of the Black–Scholes–Merton equation, which it must be, since both rest on the same no-arbitrage principle.

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Example 15.1 Pricing a call and a put

A non-dividend stock trades at \$58. Value six-month options struck at \$55 with $r = 4\%$ and $\sigma = 24\%$.

First $\sigma\sqrt{T} = 0.24\sqrt{0.5} = 0.16971$, and

$$d_1 = \frac{\ln(58/55) + (0.04 + 0.0288) \times 0.5}{0.16971} = \frac{0.053110 + 0.034400}{0.16971} = 0.5157$$

$$d_2 = 0.5157 - 0.1697 = 0.3460$$

From normal tables, $N(0.5157) = 0.69697$ and $N(0.3460) = 0.6353$. Then

$$c = 58 \times 0.69697 - 55e^{-0.02} \times 0.6353 = 40.42 - 34.25 = \$6.17$$

and for the put, $N(-d_1) = 0.30303$, $N(-d_2) = 0.3647$:

$$p = 55e^{-0.02} \times 0.3647 - 58 \times 0.30303 = 19.66 - 17.58 = \$2.08$$

Always check parity. From Section 11.4, $c - p$ must equal $S_0 - Ke^{-rT}$:

$$6.1731 - 2.0840 = 4.0891, \qquad 58 - 55e^{-0.02} = 58 - 53.9109 = 4.0891$$

They agree exactly, and they must: the formulas satisfy parity identically, because $N(d)+N(-d) = 1$ for every d. A discrepancy in a spreadsheet is an arithmetic error, not a modeling question, and this is the fastest test of any implementation.

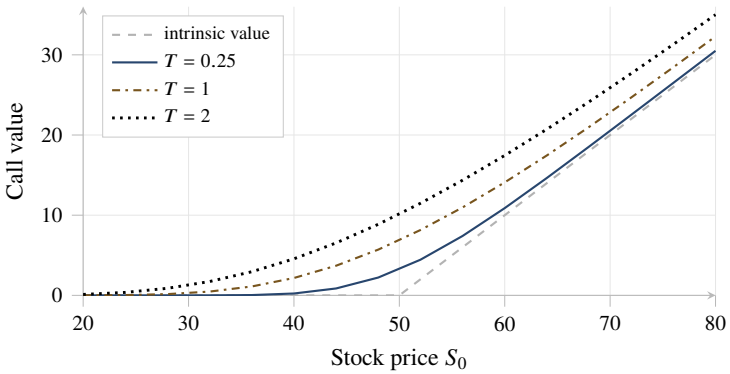


Figure 15.1: Call value against the stock price. $K = 50$, $r = 4\%$, $\sigma = 30\%$.

Figure 15.1 plots (15.2) against S_0 for three maturities. This is Figure 10.2 with a formula behind it: the curves lie above intrinsic value by the time value, the gap is widest near the money, and it shrinks toward zero as $T \to 0$. The model has not produced a new shape—it has produced a number for a shape Chapter 10 already described.

15.4 WHAT $N(d_1)$ AND $N(d_2)$ MEAN

The formula is more memorable, and more useful, once its terms are read rather than computed.

Rewrite (15.2) as

$$c = e^{-rT}\Big[\ \underbrace{S_0 e^{rT} N(d_1)}_{\text{expected stock received}} \ - \ \underbrace{K\,N(d_2)}_{\text{expected strike paid}}\ \Big]$$

Under the risk-neutral measure, the terms read as follows:

$N(d_2)$ **is the probability that the option is exercised.** It is $Q(S_T > K)$, the risk-neutral probability of finishing in the money. In Example 15.1, $N(d_2) = 0.6353$, so the market prices a 64% chance of exercise. This is not the real-world probability, for the reason Chapter 13 gave:

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Q weights states so that the stock earns r, not μ. A stock with a high expected return has a higher true probability of exercise than $N(d_2)$ suggests.

$N(d_1)$ **is the option's delta**, the hedge ratio $\partial c/\partial S$ that Sections 13.1 and 14.8 built the entire theory around. In the example it is 0.6970, so replicating the call requires holding 0.6970 shares per option written. Chapter 18 takes this as its starting point.

$S_0N(d_1)$ **is the present value of receiving the stock in the states where the option is exercised.** It is not S_0 times the exercise probability. The option is exercised only in the high-price states, those with $S_T > K$, so the stock's average value across those states exceeds its average over all states, the forward price S_0e^{rT}; $N(d_1)$ is $N(d_2)$ scaled up by that ratio, which is why $N(d_1) > N(d_2)$.

That last point resolves a persistent confusion. Only $N(d_2)$ is the exercise probability under the risk-neutral measure; $N(d_1)$ is the exercise probability under the measure that uses the stock as numeraire ([TN15]; Chapter 28).

15.5 PROPERTIES OF THE FORMULA

Any formula should be tested at the edges. Equation (15.2) passes every test, and each reproduces a result Chapter 11 obtained *without* a model.

Deep in the money. As $S_0 \to \infty$, both d_1 and $d_2 \to \infty$, so $N(d_1), N(d_2) \to 1$ and

$$c \to S_0 - Ke^{-rT}$$

The call becomes a forward: exercise is certain, so the holder is committed to buy at K.

Deep out of the money. As $S_0 \to 0$, both arguments $\to -\infty$ and $c \to 0$.

Zero volatility. As $\sigma \to 0$, the option's outcome is certain. Then $d_1, d_2 \to \pm\infty$ according to whether S_0 exceeds Ke^{-rT}, and

$$c \to \max\left(S_0 - Ke^{-rT},\, 0\right)$$

which is exactly the lower bound (11.1) of Section 11.2.

Infinite volatility. As $\sigma \to \infty$, $d_1 \to +\infty$ while $d_2 \to -\infty$, so $N(d_1) \to 1$, $N(d_2) \to 0$, and

$$c \to S_0$$

the upper bound of Section 11.2. The formula cannot be made to violate the bounds however the inputs are abused.

At expiration. As $T \to 0$, $c \to \max(S_0 - K, 0)$.

The put runs the same checks in mirror image. As $S_0 \to \infty$, both $N(-d_1)$ and $N(-d_2)$ vanish and $p \to 0$; as $S_0 \to 0$ they tend to one and $p \to Ke^{-rT} - S_0 \to Ke^{-rT}$, the put's upper bound of Section 11.2, which is also its value as $\sigma \to \infty$. As $\sigma \to 0$, $p \to \max(Ke^{-rT} - S_0, 0)$, the put's lower bound (11.2), and as $T \to 0$, $p \to \max(K - S_0, 0)$. None of these needs a separate calculation: since (15.3) follows from (15.2) by parity, each put limit is the corresponding call limit plus $Ke^{-rT} - S_0$.

The agreement with Chapter 11 is not a coincidence. Those bounds followed from no-arbitrage alone; this formula followed from no-arbitrage *plus* a distributional assumption. A model that added an assumption and thereby violated a conclusion reachable without it would be internally inconsistent. Extra assumptions may narrow a range of admissible prices; they must never move it.

Computing the normal distribution function

Equation (15.2) requires $N(x)$, which has no elementary closed form. Pricing libraries compute it as $\operatorname{erfc}(-x/\sqrt{2})/2$ from the double-precision complementary error function, to machine

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precision. By hand or in a spreadsheet a polynomial serves; a standard one, accurate to 7.5×10^{-8}, sets $\gamma = 0.2316419$ and $k = 1/(1 + \gamma x)$ for $x \geq 0$, then

$$N(x) \approx 1 - \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \left(a_1 k + a_2 k^2 + a_3 k^3 + a_4 k^4 + a_5 k^5 \right)$$

with $a_1 = 0.319381530$, $a_2 = -0.356563782$, $a_3 = 1.781477937$, $a_4 = -1.821255978$, $a_5 = 1.330274429$. For negative arguments, $N(-x) = 1 - N(x)$.

That accuracy is far beyond what pricing near the money requires (a bid–ask spread is typically five orders of magnitude wider), but not in the far wings, where $N(-d)$ is around 10^{-6}, nor when implied volatility is computed by iteration, where an error in N is divided by the option's sensitivity to volatility, which is small far from the money.

15.6 ESTIMATING VOLATILITY FROM HISTORY

Volatility must be estimated, and the two ways to do it answer different questions.

The historical estimator. Take $n + 1$ successive closing prices $S_0, S_1, \ldots, S_n$ and form the log returns $u_i = \ln(S_i / S_{i-1})$. The sample standard deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (u_i - \bar{u})^2} \tag{15.5}$$

where $\bar{u}$ is the mean of the u_i, estimates volatility per observation interval, and is annualized by multiplying by $\sqrt{252}$ for daily data, $\sqrt{52}$ for weekly, $\sqrt{12}$ for monthly.

Table 15.1: Ten daily returns.

Day	Close	u_i	Day	Close	u_i
0	50.00	—	6	51.10	+0.013793
1	50.75	+0.014889	7	51.85	+0.014570
2	50.20	−0.010897	8	51.60	−0.004833
3	51.30	+0.021676	9	52.30	+0.013475
4	51.05	−0.004885	10	52.00	−0.005753
5	50.40	−0.012814			

Example 15.2 Computing a historical volatility

From Table 15.1, $\sum u_i = 0.039221$ and $\sum u_i^2 = 0.00163889$. Then

$$s^2 = \frac{1}{9} \left[0.00163889 - \frac{(0.039221)^2}{10} \right] = 0.000165007$$

so $s = 0.012846$ per day, and

$$\sigma = 0.012846\sqrt{252} = 0.012846 \times 15.8745 = 20.4\%$$

Three cautions apply.

Ten observations is far too few. The standard error of a volatility estimate is approximately $\sigma/\sqrt{2n}$, here $20.4/\sqrt{20} = 4.6$ percentage points. A two-standard-error band runs from about 11% to 30%, which is not a usable input. Ninety to 180 observations is the usual compromise: long enough to reduce the error, short enough that volatility has not yet changed.

Trading days, not calendar days. Volatility accumulates while markets are open, not while time passes: French and Roll (1986) found the variance over a weekend far below three days' worth. So the annualization factor is $\sqrt{252}$; using 365 would overstate volatility by about 20%.

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The mean is usually dropped. Over short intervals the drift is negligible next to the noise (Example 14.3 showed it is outweighed about thirteen and a half to one over a week), so setting $\bar{u} = 0$ in (15.5), and dividing by n rather than $n - 1$ since no mean is now estimated, typically improves the estimate by removing a noisily estimated parameter. Here it gives $s = 0.012802$, changing the annualized figure by 0.1 percentage points.

What an error in volatility costs

The standard error above is easy to accept and easy to underrate; Table 15.2 prices it.

Table 15.2: Value of the call of Example 15.1 at different volatilities.

σ	Call value	Difference	Percent
14%	4.8292	−1.3440	−21.8%
19%	5.4736	−0.6995	−11.3%
24%	6.1731	—	—
29%	6.9003	+0.7272	+11.8%
34%	7.6428	+1.4697	+23.8%

The outer rows are roughly two standard errors of a ten-observation estimate on either side of 24% (the same $\pm 2\sigma/\sqrt{2n}$ band Example 15.2 computed, here about ±10 points): a 14% to 34% range for volatility corresponds to a \$4.83 to \$7.64 range for the option. The historical estimate from ten observations does not determine the price to better than about ±20%.

Two conclusions follow. Historical volatility needs far more data than a casual calculation suggests. And the relationship is close to linear over this range, which is the sensitivity Chapter 18 names vega and which Section 15.7 exploits to invert the formula.

The deeper objection to historical volatility is not statistical. It is that the estimator answers the wrong question: it measures what the stock *did*, and an option's value depends on what it *will do*.

15.7 IMPLIED VOLATILITY

The alternative inverts the problem. Take the option's market price as given and ask what volatility makes (15.2) reproduce it. That number is the **implied volatility**.

The inversion is well posed because a call's value is increasing in σ, as Section 11.1 argued without a model, and strictly so in the formula, whose vega is positive for every $\sigma > 0$; Figure 15.2 shows it for the call of Example 15.3. A strictly increasing function takes each value at most once, so an implied volatility is unique, and it exists whenever the price lies strictly between the bounds of Section 11.2. No closed form exists, so it is found numerically, which on modern hardware is instantaneous.

Figure 15.2 also shows a feature that matters in practice. Below about 5% volatility the curve is indistinguishable from the lower bound $S_0 - Ke^{-rT} = 1.149$: the call is in the money, and with so little volatility it almost never finishes out of the money, so extra volatility adds almost nothing. The function is still strictly increasing, but it is nearly flat there, so a small error in the price moves the implied volatility a long way. The put at the same strike has the same vega, so switching to it does not steepen the curve; what it changes is the price being read. Most of the in-the-money call's price is the lower bound, and only the remainder carries information about volatility, whereas the out-of-the-money put's price is all time value and usually trades on a narrower spread. That is why implied volatility at each strike is read from the out-of-the-money option, here the put, which by parity implies the same volatility as the call.

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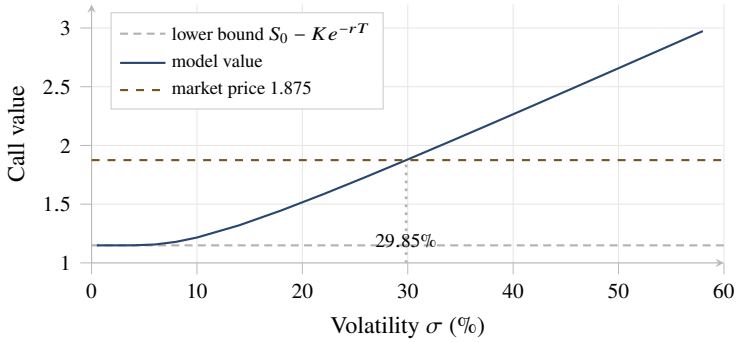


Figure 15.2: Call value against volatility, and the implied volatility. $S_0 = 21$, $K = 20$, $r = 3\%$, $T = 0.25$.

Example 15.3 Backing out an implied volatility

A three-month call on a \$21 stock, struck at \$20 with $r = 3\%$, trades at \$1.875.

Try $\sigma = 20\%$: the formula gives \$1.5148, too low. Try 30%: \$1.8807, slightly too high. Try 29%: \$1.8429. Narrowing further gives

$$\sigma_{\text{implied}} = 29.85\%$$

The market price is consistent with 29.85% volatility over the next three months. Had this stock's own history given a figure near 20%, as it did for the stock of Example 15.2, two readings would be possible: the market anticipates something the past does not contain (an earnings announcement, a pending decision), or option buyers are paying the **volatility risk premium**, the amount by which implied volatility tends to exceed the volatility later realized, which Chapter 19 documents. Usually both are true.

No one used (15.2) to compute a price here; the price was given, and the formula converted it into a volatility. That is what it is mostly used for in practice, and Section 10.6 already reported the consequence: dealers quote in volatility because volatility is the coordinate in which prices are stable.

The formula as a language. This reverses the usual account of what the formula is for. As a device for computing prices from a volatility forecast, it is unreliable, because volatility is neither constant nor reliably forecastable over an option's life. As a device for converting between prices and volatilities (a common language in which options of different strikes and maturities can be compared), it is indispensable, and its failures are informative rather than fatal.

Chapter 19 shows that implied volatility varies systematically with strike and maturity, so the model is wrong; the market uses it anyway, because a wrong but agreed transformation is still a usable quoting convention. The VIX of Chapters 19 and 22 carries the same idea to an index: one volatility number read from a strip of option prices.

15.8 USING THE FORMULA IN PRACTICE

Three conventions decide whether two people pricing the same option agree.

Which interest rate? The model says "the" risk-free rate, and Section 5.15 explained why no single such rate exists: which one a participant faces depends on who they are and what they can pledge. In practice, the rate used is the one at which the hedge can be financed over the option's life, which for a dealer's OTC trade, cleared or collateralized under a CSA, is the collateral rate: the OIS rate of Chapter 4. For listed options, desks commonly take r and q

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(with any borrow cost) jointly from the implied forward of Section 11.6 and the box-spread rate of Section 11.7. Using a government yield instead is a small error for short-dated equity options and a large one for long-dated rate derivatives, which is why Chapters 7 and 8 built the discounting apparatus they did.

Which maturity? Section 15.6 annualized volatility using *trading* days, and the two clocks must be kept apart: variance accrues on trading days, interest on calendar days. An option expiring in 30 calendar days has standard deviation $\sigma\sqrt{21/252}$ over its life but discount factor $e^{-30r/365}$, so d_1 carries rT_{cal} beside $\sigma^2 T_{\text{trd}}/2$. Mixing the conventions is a common implementation error. On average the two agree, since $252/365 \approx 21/30$, but for an option with a week or two to run the placement of weekends and holidays inside its life moves the variance by several percent. Some desks go further and weight days unequally (Chapter 23's business time: a day containing an earnings announcement carries more variance than a quiet Tuesday), which is a refinement of the same idea.

Which volatility? This is the substantive one. The formula assumes a single σ governs the stock, so any option on it should use the same number. The market disagrees: implied volatilities differ across strikes and maturities, as Section 15.11 notes and Chapter 19 documents.

The working resolution is to abandon the pretense of one volatility and use, for each option, the implied volatility of that option, interpolated from the traded surface if the exact strike and maturity are not quoted. The formula is then a quoting convention, strike by strike, rather than a model of the stock. A desk that prices a book with a single volatility is not using Black–Scholes conservatively; it is using a model the market has already rejected.

15.9 DIVIDENDS

Assumption 4 can be relaxed in two ways, according to how dividends are described.

A continuous dividend yield. If the asset pays a known yield q, its holder receives q and the option holder does not. Section 5.4 showed the effect: the asset behaves like one worth $S_0 e^{-qT}$ today. Replacing S_0 by $S_0 e^{-qT}$ throughout gives

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2) \tag{15.6}$$

$$d_1 = \frac{\ln(S_0/K) + \left(r - q + \sigma^2/2\right) T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

This single substitution covers stock indices (q = dividend yield), currencies ($q = r_f$, the foreign rate), and futures ($q = r$), and Chapter 16 develops each; Example 15.4 applies it to an index.

The differential equation changes as well, and the change explains the substitution. The shares in the hedge portfolio now pay dividends of

$$qS\frac{\partial f}{\partial S}\,dt$$

over each interval, which the riskless return must include, so the argument of Section 15.2 gives

$$\frac{\partial f}{\partial t} + (r - q)S\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf \tag{15.7}$$

The asset's risk-neutral drift is $r - q$ while discounting is still at r: the option holder forgoes the yield, so the price the option sees grows more slowly than the riskless rate. Solving (15.7)

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with the call's boundary condition gives (15.6), and setting $q = r$ removes the first-derivative term altogether, which is the form Chapter 16 uses for options on futures.

Example 15.4 An index option

An index stands at 1,500 with a 2.5% dividend yield. Value a six-month call struck at 1,550 with $r = 4\%$ and $\sigma = 18\%$.

$$d_1 = \frac{\ln(1500/1550) + (0.04 - 0.025 + 0.0162) \times 0.5}{0.18\sqrt{0.5}} = -0.135056, \qquad d_2 = -0.262335$$

$$c = 1500e^{-0.0125} \times 0.446284 - 1550e^{-0.02} \times 0.396532 = 661.11 - 602.45 = 58.66$$

At the \$100 multiplier of Section 10.3, one contract costs \$5,866.

Here $N(d_2) = 0.3965$: the market prices a 40% risk-neutral chance that the index gains more than 3.3% in six months.

Known cash dividends. For an individual stock the dividend is a known amount on a known date, not a yield. The standard treatment subtracts the present value of the dividends payable during the option's life and applies (15.2) to what remains (Example 15.5).

Example 15.5 A call on a dividend-paying stock

A stock trades at \$50 and will pay \$0.80 in two months and \$0.80 in five months. Value a six-month call struck at \$50 with $r = 4\%$ and $\sigma = 30\%$.

The present value of the dividends is

$$0.80e^{-0.04(2/12)} + 0.80e^{-0.04(5/12)} = 0.7947 + 0.7868 = \$1.5815$$

so the adjusted stock price is $S^* = 50 - 1.5815 = \$48.4185$. Applying (15.2) to S^* gives

$$c = \$3.83$$

against \$4.70 for an otherwise identical option on a stock paying nothing, so the dividends have cost the call 18.5% of its value.

The volatility in this calculation is that of S^*, not of S, and the two differ because S^* is a smaller number carrying the same absolute risk. For a modest yield the difference is negligible; for a high yield over a long horizon it is not. [TN15] quantifies it.

15.10 AMERICAN OPTIONS

Equations (15.2) and (15.3) price *European* options. Chapter 11 established when that matters.

American calls on non-dividend stocks need nothing: Section 11.8 proved early exercise is never optimal, so $C = c$ and (15.2) is exact.

American puts have no closed-form solution, and this is not a gap awaiting a cleverer derivation—it is a consequence of the problem's structure. The holder's exercise decision must be made at every date before expiration, so the boundary condition of Section 15.2 is not fixed at T but applies along a **free boundary** (the critical stock price at each date below which immediate exercise is optimal), whose location is itself part of the answer. Equation (15.1) still holds; what is missing is a closed form for where it stops holding.

Three responses are used, in ascending order of cost.

Use the European value as a lower bound. Since $P \geq p$ always (Section 11.9), the formula gives a floor. How slack the floor is depends on how deep in the money the option is and on the level of rates. The two-year put of Section 13.9's convergence figure is worth 6.7601 as a European, the limit the tree settles to, and 7.4719 as an American on a fine tree, an early-exercise premium of 10.5%, far too large to ignore. For a one-month out-of-the-money put on the same

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stock it is under a percent.

Use an analytic approximation. Several exist. The best known, the Barone-Adesi–Whaley quadratic approximation, drops a small term from the early-exercise premium's equation so that it solves in closed form, then finds the boundary by iteration. Such approximations are fast and typically accurate to a few tenths of a percent, inside the bid–ask spread for most listed options.

Use a lattice. Chapter 13's backward induction handles the free boundary exactly by testing exercise at every node, and Chapter 20 makes it efficient. This, or the finite-difference grid of Chapter 20, is what most dealers' pricing systems use, and a large part of why the formula has not retired the tree.

American calls on dividend-paying stocks may, in a frictionless market, be exercised early, but only immediately before an ex-dividend date (Section 11.10). That restriction permits a serviceable approximation, worked through in Example 15.6.

Example 15.6 Black's approximation

A stock trades at \$52 and pays a \$1.50 dividend in five months. Value an American call struck at \$50 expiring in six months, with $r = 4\%$ and $\sigma = 30\%$.

First check whether early exercise is even possible, using condition (11.13):

$$D = 1.50 \quad \text{against} \quad K\left(1 - e^{-r(T-t_1)}\right) = 50\left(1 - e^{-0.04/12}\right) = \$0.17$$

where $t_1 = 5/12$ is the ex-dividend date. The condition is comfortably satisfied, so exercise just before the ex-dividend date may be optimal.

Black's approximation values the American call as the greater of two European calls: one expiring at T on the dividend-adjusted price, and one expiring just before the ex-dividend date at t_1 on the unadjusted price.

European to T. Subtracting the dividend's present value, $\mathrm{PV}(D) = 1.50e^{-0.04(5/12)} = \$1.4752$, gives $S^* = \$50.5248$, and (15.2) with $T = 0.5$ gives \$5.00.

European to $t_1 = 5/12$. Here the option is assumed exercised before the dividend, so no adjustment applies. With $S_0 = 52$ and $T = 5/12$, (15.2) gives \$5.48.

Black's approximation takes the larger:

$$C \approx \max(5.00,\ 5.48) = \$5.48$$

so, of the two policies, exercising just before the ex-dividend date is worth more than holding to expiration.

The approximation is crude, and it always understates: it chooses between the two policies today, while the holder chooses at the dividend date after seeing the price; that option to decide later is what it omits. Its merit is that it requires two evaluations of a formula rather than a lattice, and with one dividend it is usually within a few cents of the tree value.

15.11 WHAT THE MODEL GETS WRONG

The formula was published in 1973, the options market adopted it within two or three years, and it has been known to be wrong since at least 1987. Both facts are worth taking seriously.

Volatility is not constant across strikes. If the model were right, every option on a stock would imply the same σ. They do not, and the failure can be measured on the book's own data.

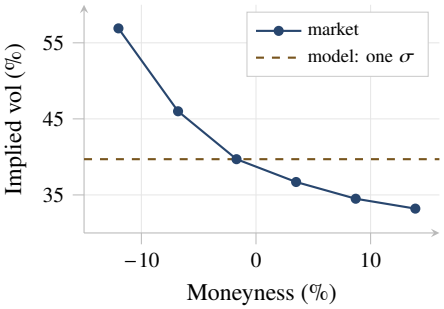
Table 15.3 inverts (15.2) across the option chain of Table 10.2, using the \$1.00 dividend that Section 11.6 extracted from the same quotes.

The implied volatility falls monotonically from 56.9% to 33.2% as the strike rises. This is a single stock, on a single day, at a single maturity, and the model requires all six numbers to be identical. They differ by a factor of 1.7. The calls are American, so part of each price, most at the lowest strikes, is early-exercise premium; inverting a European formula folds it into the implied volatility, which a first reading accepts and Chapter 19 refines.

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Table 15.3: Call implied volatilities from the chain of Table 10.2. $S_0 = 48.30$, $r = 5\%$, discounting over 45 calendar days, variance over 31 trading days; moneyness is $K/S_0 - 1$.

Strike	Moneyness	Call mid	Implied vol
42.50	−12.0%	6.650	56.9%
45.00	−6.8%	4.425	46.0%
47.50	−1.7%	2.670	39.7%
50.00	+3.5%	1.475	36.7%
52.50	+8.7%	0.710	34.5%
55.00	+13.9%	0.305	33.2%



The shape (high implied volatility at low strikes, falling as the strike rises) is the equity skew. It has been a stable feature of index and single-stock options since the crash of October 1987, and it says something specific: the market assigns more probability to large downward moves than the lognormal distribution of (14.11) allows. Chapter 19 is devoted to reading it.

Put–call parity (Section 11.4) holds identically for the Black–Scholes–Merton formulas, so a call and a put at the same strike and maturity must imply the same volatility. If European options do not, the cause is a data problem (a stale quote, a wrong dividend or borrow-fee assumption, a mismatched timestamp) and not a market phenomenon; on American-style listed options a residual gap is the early-exercise premium, which Section 11.6 already met in this chain. Checking it is the first test on any option dataset.

Volatility is not constant across time either. Implied volatility varies with maturity, and that term structure (the pattern of implied volatility across maturities) itself moves over time.

Returns are not lognormal. Real returns have fatter tails than the lognormal distribution behind (15.2), and jumps that no continuous path allows. The skew of Table 15.3 is that disagreement showing up in prices: low strikes are priced for a fatter lower tail than the model allows.

The frictionless assumptions (2, 3, and 7) fail as Section 15.1 noted. The model remains the market's language for the reason Section 15.7 gave: it is used not as a price generator but as an invertible map between prices and a single number, and for that purpose being wrong in a stable, well-understood way is tolerable. The volatility surface of Chapter 19 is not a correction to the model—it is a catalog of exactly how the model is wrong, quoted in the model's own units.

Case Study 15.1 The formula that arrived with the market

The Chicago Board Options Exchange opened on April 26, 1973. Black and Scholes's paper appeared in the *Journal of Political Economy* that May, having been rejected twice before publication; Merton's paper, developed alongside theirs, appeared in the *Bell Journal* the same spring. A market and its pricing model were born within weeks of each other.

Adoption was extraordinarily fast for an academic result. Traders on the CBOE floor were using the formula within two or three years, from about 1975 first on Black's printed sheets of model values and then on a handheld calculator sold pre-programmed with it. By the end of the decade it was the market's common language, and Scholes and Merton received the Nobel Prize in 1997. Black had died in 1995, and the prize is not awarded posthumously.

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What makes the episode instructive is that the model's assumptions were violated on the day it was published, and everyone involved knew it. Stocks pay dividends, volatility moves, markets close overnight, borrowing is not at the risk-free rate. Black said so himself, at length, in a 1988 essay titled "The Holes in Black–Scholes"—a list of the model's defects written by one of its authors, which remains accurate today.

The formula spread anyway, as a quoting convention rather than a pricing model, and no better convention has displaced it in fifty years.

15.12 WHERE THIS CAME FROM

Bachelier (1900) is the beginning, and it is remarkable twice over: he modeled the price itself, not its logarithm, as Brownian motion (the arithmetic Wiener process $dS = \sigma\, dW$ of Section 14.3) five years before Einstein described the physics, and he derived option values from it. The normal model's flaw (prices can go negative) kept it out of use for most of the twentieth century, and it returned in the 2010s when negative rates made it the natural choice for rate options, and again in April 2020, when WTI futures settled below zero and CME moved its crude-oil options to it for four months.

The intervening attempts show what was missing. Samuelson (1965), adopting the geometric Brownian motion Osborne had proposed, got the distribution right, but priced the option by discounting its expected payoff at a rate that had to be assumed. No one could say what that rate should be, and that single gap is what Black and Scholes (1973) closed: a continuously rebalanced hedge removes the risk, so no risk premium is needed and no view about the stock is required. Merton (1973) reached the same formula by a different route in the same year and extended it to dividends and to stochastic rates.

Cox and Ross (1976) then extracted the principle from the formula. If the hedge argument works, the option can be valued as though every asset drifted at the risk-free rate—not because investors are risk-neutral but because the answer does not depend on whether they are. Cox, Ross and Rubinstein (1979), with Rendleman and Bartter (1979) independently in the same year, turned that principle into Chapter 13's tree, and Harrison and Kreps (1979) with Harrison and Pliska (1981) gave the general statement Chapter 28 uses: no arbitrage holds if and only if an equivalent martingale measure exists, a risk-neutral measure of Chapter 13's kind under which every discounted asset price has zero drift.

SUMMARY

The Black–Scholes–Merton equation follows from Chapter 14's observation that a portfolio short one derivative and long $\partial f/\partial S$ shares has no random component. Setting its return to r gives $\partial f/\partial t + rS\,\partial f/\partial S + \frac{1}{2}\sigma^2 S^2\,\partial^2 f/\partial S^2 = rf$. The stock's expected return μ cancels, which is Chapter 13's principle in continuous time, and the equation governs every derivative whose value depends only on S and t: the instrument is the boundary condition, not the equation. A forward contract satisfies it identically, which confirms that Chapter 5 and this chapter rest on the same principle.

Solving it for a European call gives $c = S_0 N(d_1) - Ke^{-rT}N(d_2)$, with the put following from parity. The formulas satisfy put–call parity identically, which is the fastest test of any implementation.

Of the two normal terms, $N(d_2)$ is the risk-neutral probability of exercise, not the real-world one. $N(d_1)$ is the delta, and $S_0 N(d_1)$ is the present value of receiving the stock in the states where the option is exercised.

Volatility is the one input that must be estimated. The historical estimator uses the sample standard deviation of log returns, annualized by $\sqrt{252}$; over a short sample its standard error

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is large, and it answers the wrong question, measuring what the stock did rather than what it will do. Implied volatility inverts the formula against a market price. The inversion is unique because option value is strictly increasing in volatility.

The formula's real role is as a language. It converts prices into a single comparable number, which is why dealers quote in volatility and why a model known to be wrong remains universal.

A continuous dividend yield is handled by replacing S_0 with $S_0 e^{-qT}$ (equivalently, the equation's drift term becomes $(r - q)S\,\partial f/\partial S$), which also covers indices, currencies, and futures. Known cash dividends are subtracted at present value.

American calls on non-dividend stocks are priced exactly by the European formula. American puts need a lattice or an analytic approximation. American calls on dividend-paying stocks can be valued by Black's approximation, the greater of two European values.

The model is wrong in documented ways: volatility varies with strike and maturity, returns have fat tails and jumps, and the frictionless-market assumptions fail. Measured on the book's own option chain, implied volatility differs widely across strikes. Chapter 19 catalogs the first of these in the model's own units.

FURTHER READING

[TN15] derives (15.2) from the risk-neutral expectation and shows the equivalence with solving the differential equation. The original papers, Black and Scholes (1973) and Merton (1973), are worth reading for how differently the two derive the same result: Black and Scholes through a hedged portfolio and the heat equation; Merton through a self-financing replicating portfolio (one that needs no cash added or withdrawn after it is set up) under weaker assumptions; the risk-neutral route this chapter prefers is Cox and Ross (1976). Black's 1988 essay "The Holes in Black–Scholes" is a short account of the model's failures by one of its authors, and its list still applies nearly four decades on.

PRACTICE QUESTIONS

15.1 A non-dividend stock trades at \$30. Value a three-month European call struck at \$29 with $r = 5\%$ and $\sigma = 25\%$. Then value the put and verify put–call parity.

15.2 Explain why the stock's expected return does not appear in (15.1), and state which chapter established the same result in a discrete setting.

15.3 In Example 15.1, state the risk-neutral probability that the option is exercised and explain why this is not the probability an investor should use when forecasting.

15.4 Compute the historical volatility from the last six observations of Table 15.1 and compare it with the estimate from all ten. What does the difference tell you about the reliability of short samples?

15.5 A call on a \$60 stock struck at \$60 with six months to run and $r = 4\%$ trades at \$5.20. Estimate the implied volatility to the nearest half percent by trial and error.

15.6 An index at 2,000 with a 2% dividend yield has three-month calls struck at 2,050 trading at 42. With $r = 3\%$, find the implied volatility.

15.7 A stock at \$40 pays a \$0.60 dividend in three months. Value a five-month European call struck at \$40 with $r = 5\%$ and $\sigma = 28\%$.

15.8 Explain why the European formula prices an American call on a non-dividend stock exactly, but not an American put.

15.9 Verify the limiting behavior of (15.2) as $\sigma \to \infty$ and confirm that it reproduces the upper bound of Section 11.2. Do the same as $\sigma \to 0$.

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15.10 Show that a forward contract satisfies (15.1), and explain what this demonstrates about the relationship between Chapter 5 and this chapter.

15.11 Using Table 15.3, state by what factor the implied volatilities differ across strikes and explain in one sentence what the model requires them to be.

FURTHER QUESTIONS

15.12 Show algebraically that (15.2) and (15.3) satisfy put–call parity for all parameter values, using $N(d) + N(-d) = 1$. Then explain why this makes parity useless as a test of whether the *model* is correct, though useful as a test of whether an *implementation* is.

15.13 Section 15.4 states that $N(d_1) > N(d_2)$ always. Prove this from (15.4), and give the economic reason in one sentence.

15.14 Example 15.3 found an implied volatility of 29.85%; suppose the same stock's own history gave 20.4%, as Example 15.2's did. Construct two distinct explanations consistent with both numbers, describe an observation that would distinguish them, and state which is more likely if the gap persists across all strikes and maturities.

15.15 Black's approximation always understates the American call value. Explain why, identify the choice available to the holder that it omits, and construct a case in which the understatement would be largest.

15.16 Example 15.5 notes that the volatility in the dividend adjustment refers to S^* rather than S. Quantify this: for a stock with a 4% annual yield and a five-year option, estimate the proportional difference between σ_{S^*} and σ_S, and state whether ignoring it overstates or understates the option value.

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